

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **Python**-based data pipelines for ETL processes on **AWS S3**, processing approximately **1.5TB** of raw data weekly.
- Developed and maintained **Java Spring Boot** REST APIs for internal microservices, handling over **5,000 requests per minute** during peak hours.
- Automated production monitoring and alerting using **Splunk** queries and **Python scripts**, reducing critical incident response time by **18%** for enterprise applications.
- Developed and deployed **machine learning models** using **Scikit-Learn** for anomaly detection on time-series telemetry data, achieving **0.88 AUC** on validation sets.
- Designed and managed **large-scale data models** and schemas within **PostgreSQL** to support AI initiatives and operational forecasting.
- Containerized AI inference services using **Docker** and deployed them to **Kubernetes** clusters, enabling scalable and resilient production AI systems.
- Collaborated with engineering teams to integrate **AI capabilities** into existing systems via APIs, ensuring smooth transition into production environments.

SKILLS

Languages – Python, Java, SQL, JavaScript, TypeScript

Frameworks/Libraries – Spring Boot, Scikit-Learn, PyTorch, TensorFlow, Node.js, NestJS

Cloud/DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Data/ML – Time-series analysis, Anomaly Detection, Predictive Maintenance, Data Modeling, ETL, Machine Learning, AI Deployment

Tools – Splunk, Dynatrace, Jira, Confluence

PROJECTS

Real-time Anomaly Detection System

- Engineered a **Python** pipeline to ingest and process IoT-style telemetry data, transforming raw sensor readings into features for ML analysis.
- Trained and evaluated multiple **machine learning models** (including LSTM and Isolation Forest) on a **10M-row time-series dataset** for anomaly detection, achieving a **F1-score of 0.79**.
- Deployed the chosen anomaly detection model as a **FastAPI** service within a **Docker** container, integrated with a **PostgreSQL** database for storing detected anomalies.

Operational Forecasting Dashboard with AI Insights

- Developed a backend service in **Java Spring Boot** to aggregate operational metrics and expose them via REST APIs for frontend consumption.
- Implemented a predictive model using **Scikit-Learn** to forecast system load for the next 24 hours based on historical time-series data, reducing unexpected capacity issues by **15%**.
- Collaborated with business analysts to define data requirements and integrate AI-generated forecasts into an operational monitoring dashboard.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)